

CLIMATE & EXTREME WEATHER ANALYTICS DASHBOARD

A Metabase BI Solution Using
Météo-France Open Data



Presented By

Chifa CHOUCANE

Cyrine CHAMMEM

Yasmine BEN-HAJ-SALAH

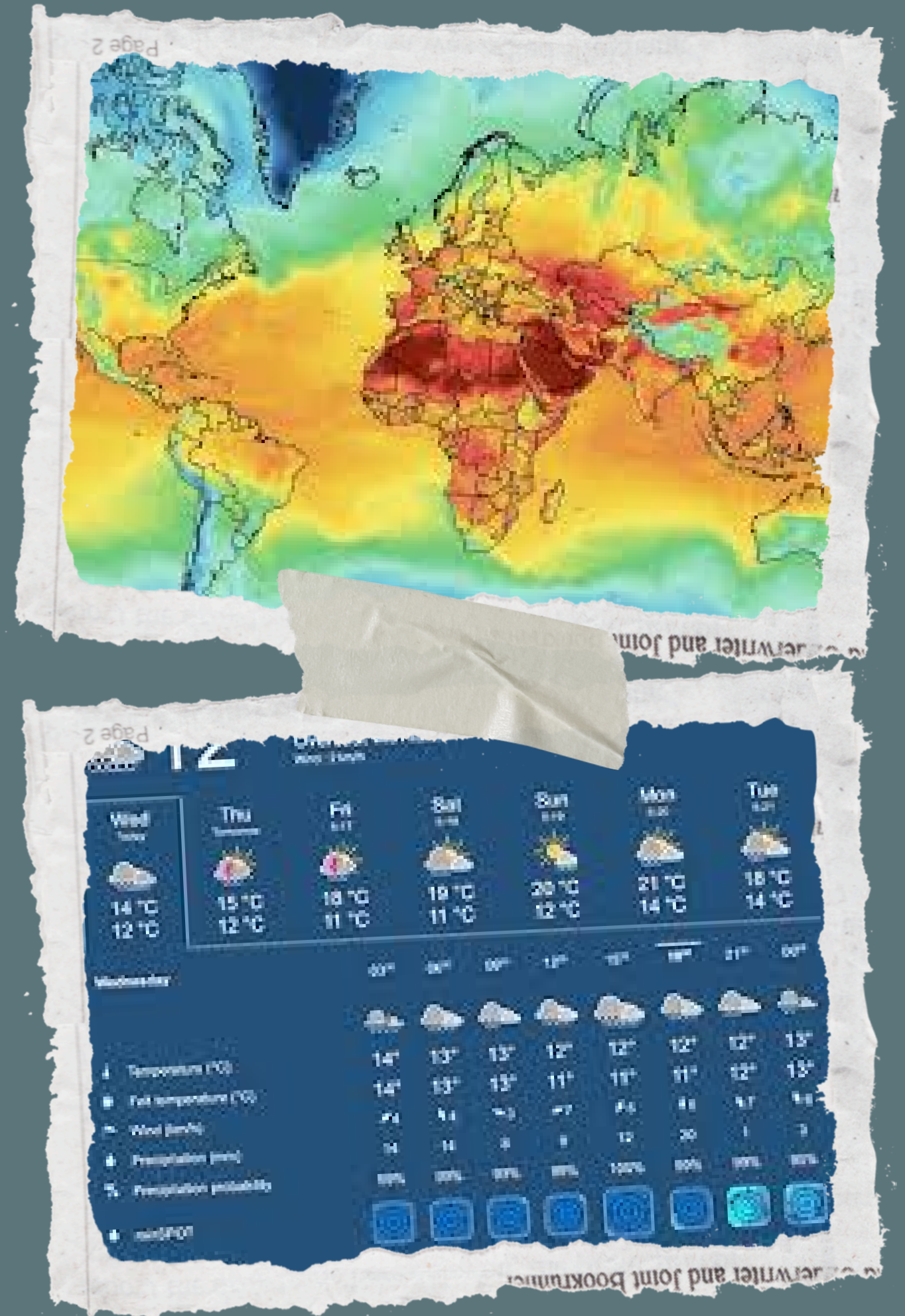
Tools: Metabase | PostgreSQL | Docker | Python



HOW CAN WE TURN RAW WEATHER DATA INTO ACTIONABLE INSIGHTS?

Raw weather data is complex and difficult to analyze directly.

Our project transforms Météo-France open data into an interactive Metabase dashboard to monitor climate trends and identify extreme weather events.





PROJECT OBJECTIVES

Our goal is to build an interactive BI dashboard that helps users explore and understand Météo-France weather data. The dashboard allows users to:

Analyze

temperature, rainfall and wind

Monitor

climate trends over time

Compare

weather stations

Identify

extreme weather events

Support

decision-making

DATA SOURCES

Main source
Météo-France OpenData

Dataset
Daily climatological weather data

Raw file
Q_01_previous-1950-2024_RR-T-Vent.csv



1

Station
num_poste
nom_usuel
lat, lon
alti

2

Date
aaaammjj

3

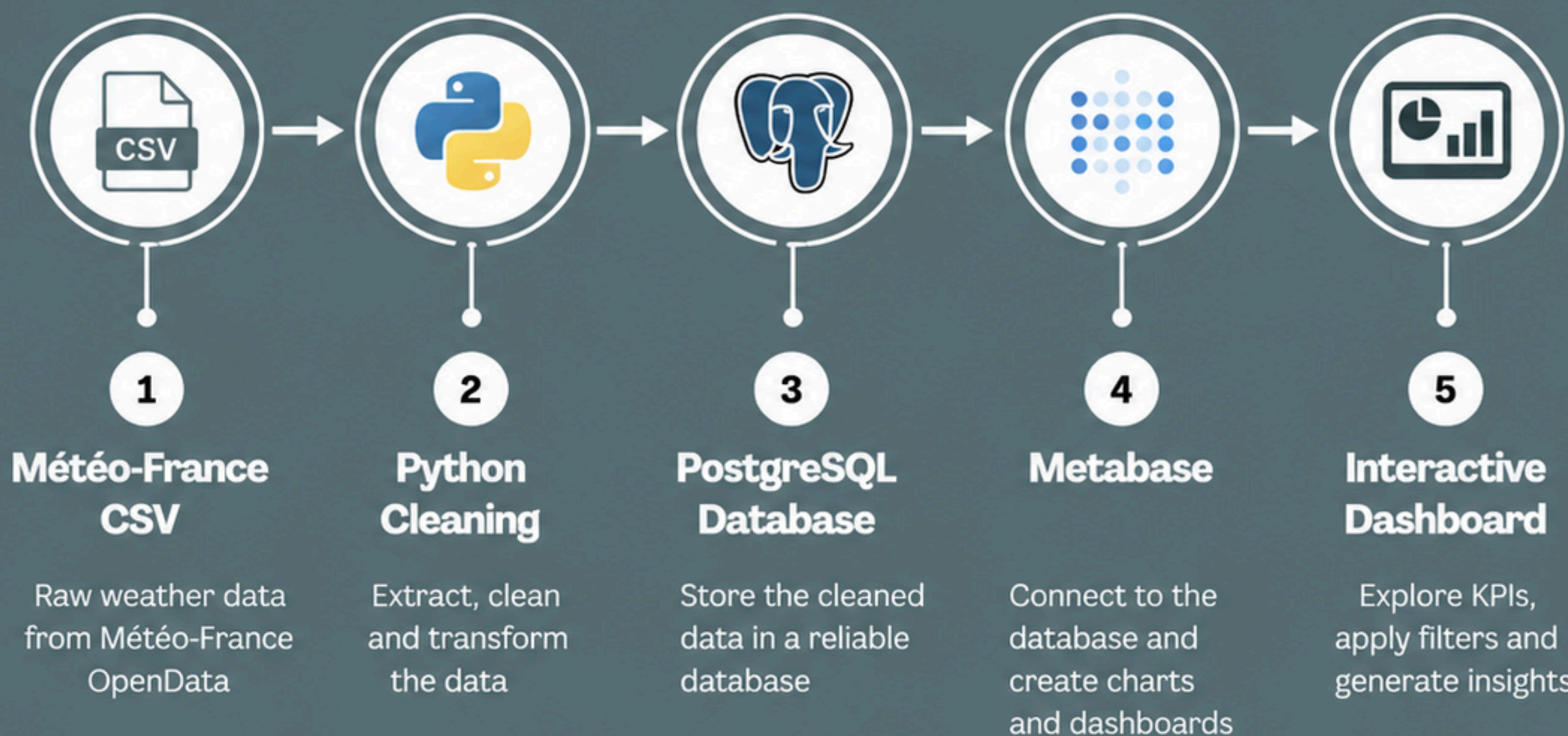
Temperature
tn
tx
tm

4

Rainfall & Wind
rr
ffm
fxy

TECHNICAL ARCHITECTURE

A complete data pipeline
built to transform raw
weather data into
interactive visual insights.



DATA PREPARATION

Raw weather data was cleaned and transformed before being loaded into PostgreSQL.



1

Load raw CSV file

Import the raw dataset from Météo-France OpenData.



2

Select useful variables

Keep the relevant columns for analysis.



3

Clean and format data

Standardize formats and remove unnecessary values.



4

Convert numerical values

Ensure the correct data types for accurate analysis.



5

Handle missing values

Check and treat missing or inconsistent data.



6

Save and load into PostgreSQL

Save the clean dataset and load it into PostgreSQL.



Output

clean_meteo.csv



Result

No missing values after cleaning

DASHBOARD OVERVIEW

Our Metabase dashboard is organized into five analytical sections to guide users from a global overview to detailed weather insights.



1

KPI Overview

2

Climate Trends

3

Weather Analysis

4

Geographic & Station Analysis

5

Extreme Events

STORYTELLING APPROACH

Our dashboard follows a progressive analytical story to transform data into actionable insights.



1

Start with global indicators

Understand the overall weather situation



2

Analyze trends over time

Compare temperature stability and rainfall variability



3

Identify patterns

Explore correlations and weather condition distribution



4

Compare weather stations

Identify stations exposed to heat, rainfall or strong wind



5

Detect extreme situations

Highlight unusual events and extract actionable insights



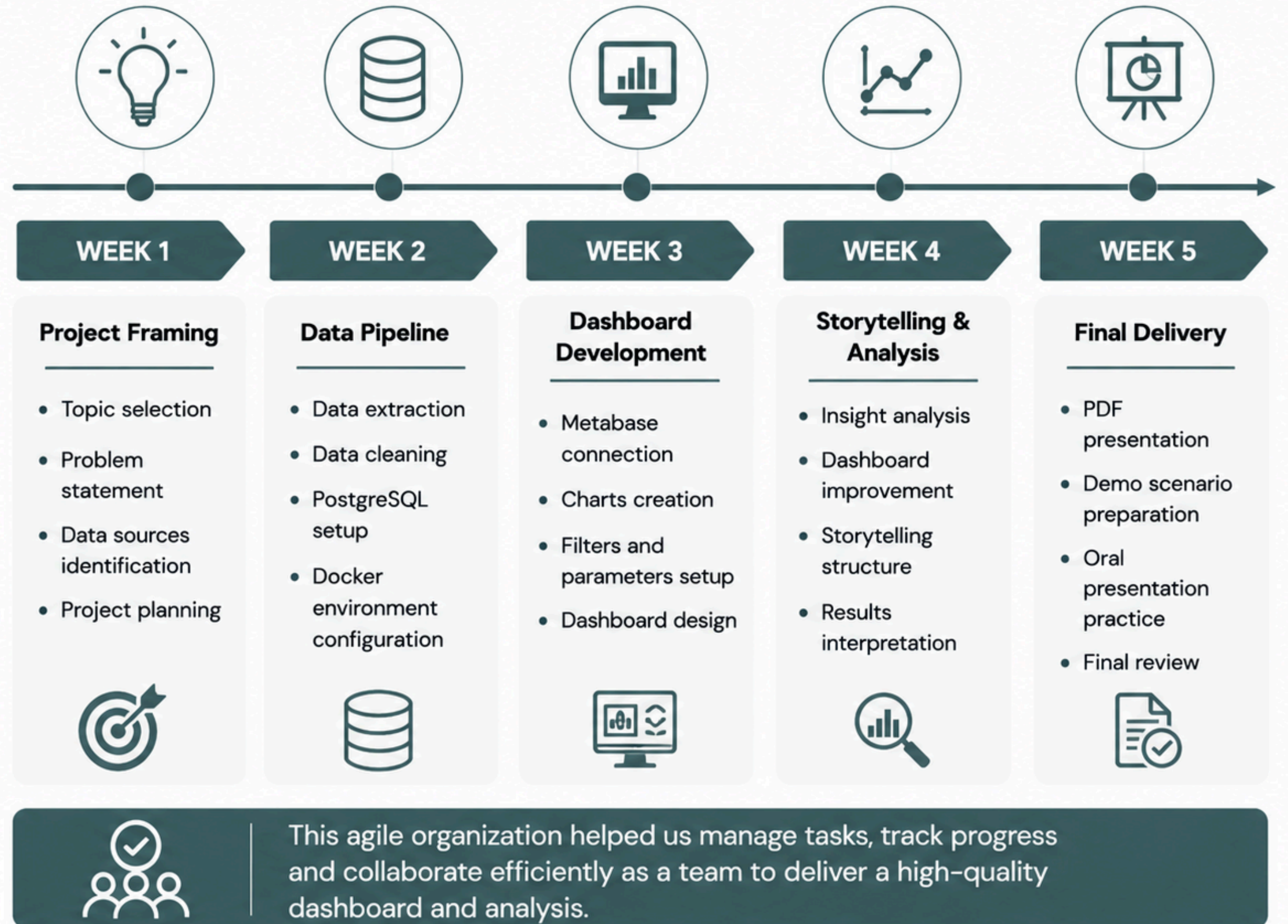
This flow helps users move from descriptive indicators to deeper weather analysis and better decision-making.

AGILE ORGANIZATION

5-week project organized into weekly sprints

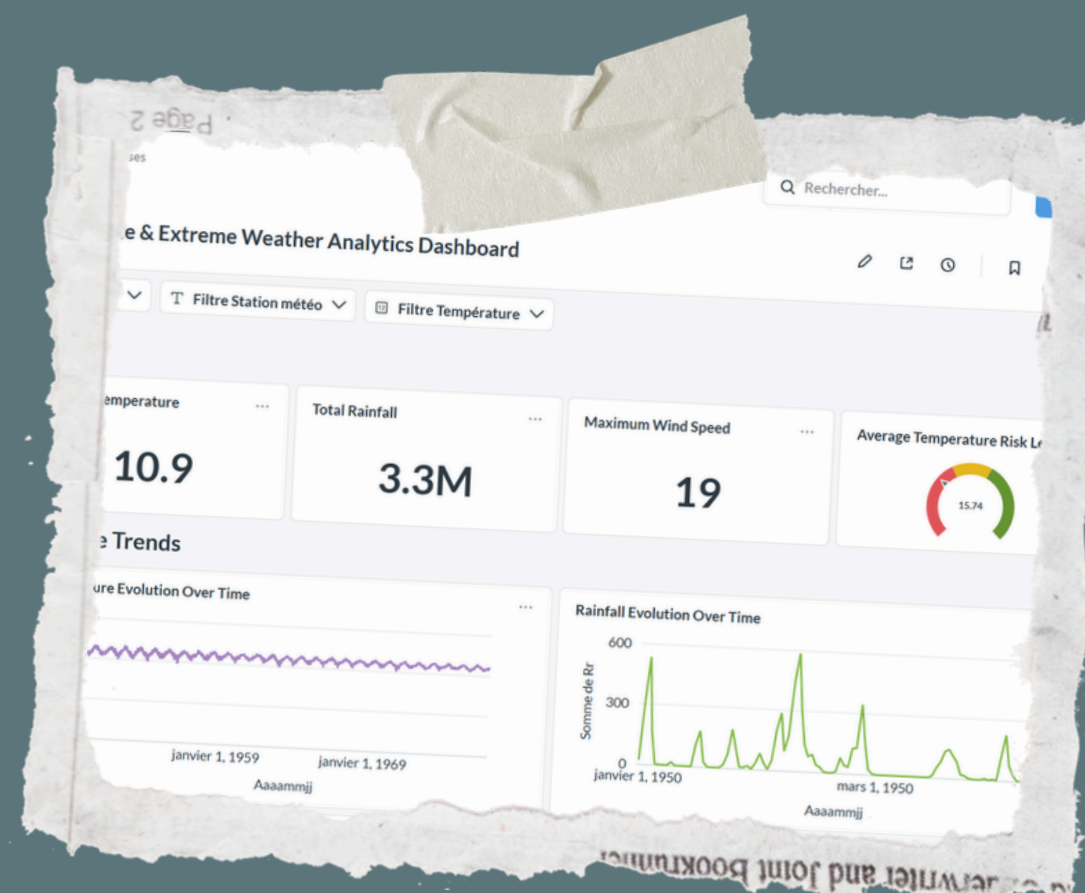


We followed an iterative and collaborative approach with weekly goals, continuous progress tracking and regular validation.





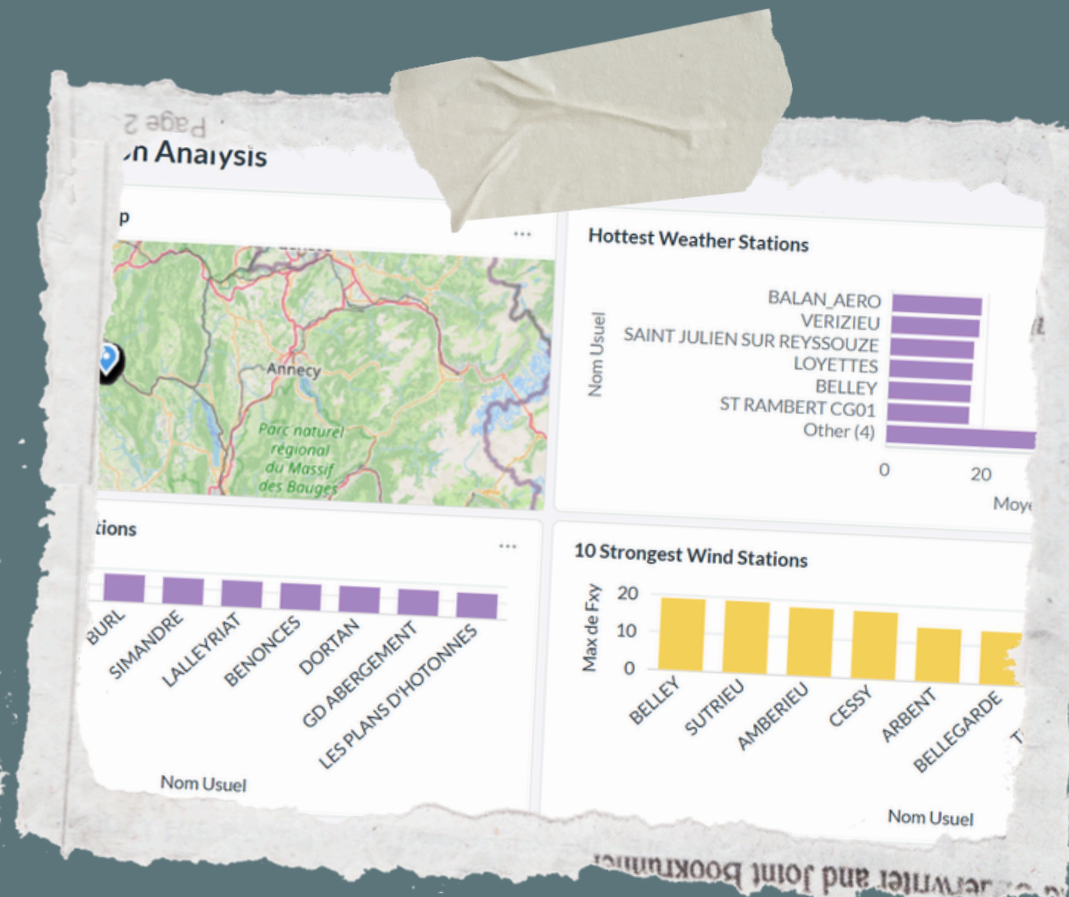
DASHBOARD DEMONSTRATION Storytelling & Analysis



Global Overview



Weather Analysis



Geographic & Station
Analysis

Conclusion

This project shows how open weather data can be transformed into useful insights through a BI solution.

By combining Météo-France OpenData, PostgreSQL, Docker and Metabase, we created an interactive dashboard to analyze climate trends and extreme weather events.

